

llmXive Automated Review of SWE-Explore: Benchmarking How Coding Agents Explore Repositories

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: [qwen.qwen3.5-122b](#)

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper makes several strong factual claims regarding the novelty of the benchmark and the statistical validity of its results. While the overall methodology is sound, there are critical factual errors regarding the model versions cited and potential inaccuracies in the comparative benchmark table that require correction.

First, in Section 3.2 (“Ground-Truth Annotation”) and the Abstract, the authors list “GPT-5.4”, “Gemini-3-Pro”, “Sonnet-4.6”, “GLM-5.1”, and “Kimi-K2.6” as the source models used to generate the successful trajectories for ground truth. As of the paper’s stated publication context (2026), these specific version numbers (e.g., GPT-5.4, Sonnet-4.6) do not correspond to any publicly released or known model versions (current generations are GPT-4o, Claude 3.5/4, Gemini 1.5, etc.). This appears to be a hallucination or a placeholder error. If these are hypothetical future models, the claim that the ground truth is derived from “successful agent trajectories” of these specific models is factually unsupported. The authors must replace these with actual, existing model versions (e.g., GPT-4o, Claude 3.5 Sonnet) or clarify the nature of these models to ensure the claim of “trajectory-grounded supervision” is valid.

Second, the comparative claim in Table 1 regarding “Trajectory-Grounded GT” requires verification. The table marks ContextBench (li2026contextbench) as lacking this feature. However, the Related Work section describes ContextBench as introducing “human-annotated gold contexts and scores retrieval over agent trajectories.” If ContextBench’s gold contexts are derived from or validated against agent trajectories, the distinction made in Table 1 might be semantically imprecise. The authors should verify the exact methodology of ContextBench to ensure the claim that SWE-Explore is the *only* benchmark with trajectory-grounded ground truth is accurate.

Finally, regarding the statistical claim in the Abstract and Section 5.2: the paper states a correlation of $r=0.950$ on $n=150$ instances. While the authors mention adding confidence intervals, the text does not explicitly report the p-value for this correlation. Given the high correlation, the claim that metrics “strongly track” repair behavior is plausible, but explicitly stating the statistical significance ($p < 0.05$) in the text would strengthen the factual support for this conclusion.

These issues are primarily factual accuracy concerns regarding the existence of cited models and the precision of comparative claims. Correcting the model names is essential for the scientific validity of the ground-truth construction.

Action items.

- **[science]** Section 3.2 lists ‘GPT-5.4’, ‘Gemini-3-Pro’, and ‘Sonnet-4.6’ as source models for ground truth. These versions do not exist. Correct to actual existing models (e.g., GPT-4o) to validate the trajectory-grounded claim.
- **[science]** Table 1 claims SWE-Explore is the only benchmark with ‘Trajectory-Grounded GT’. ContextBench (li2026contextbench) uses agent trajectories for gold contexts. Verify if this distinction is accurate or if the table entry is incorrect.
- **[writing]** Abstract claims $r=0.950$ correlation on $n=150$ instances. Explicitly state the p-value in the text to fully support the statistical significance of the ‘strong tracking’ claim.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

The figure clearly displays the requested data with a shared legend, but the caption contains a text placeholder error regarding the context degradation definition, and the drastic difference in y-axis scales between the easier and harder subsets is not explicitly highlighted.

Figure 2

Figure 2 provides a clear and comprehensive overview of the SWE-Explore framework, effectively illustrating the workflow from trajectory extraction to evaluation. The diagram is well-structured with distinct sections for benchmark construction and validation, and the caption accurately reflects the visual content.

Figure 3

Figure 3 effectively visualizes the SWE-Explore instance structure, but the terminology in the legends ('core' vs 'core hit') is inconsistent with the caption's definition of 'core span' and lacks a clear definition for the 'hit' label.

Figure 4

The figure is a clear donut chart that accurately visualizes the language distribution across the 848 instances mentioned in the caption. All segments are labeled with the language name and count, and the total is explicitly stated in the center.

Figure 5

Figure 5 effectively visualizes the paper's motivation by contrasting the 'Existing Benchmark' (which relies on a single end-to-end 'Resolve Rate' metric) with the 'SWE-Explore Benchmark' (which isolates 'Exploration Quality' for direct evaluation). The diagram is clear, well-labeled, and the caption accurately describes the conceptual shift from conflated metrics to isolated evaluation targets.

Figure 6

Figure 6 is critically flawed because it lacks a descriptive caption, appearing to be a duplicate of Figure 4. While the data sums correctly, the visualization of small categories is cluttered.

Action items.

- **[writing]** Figure 1: The caption contains a placeholder error ('to either % of R_core') instead of specifying the actual degradation percentages or fractions corresponding to the x-axis values.
- **[science]** Figure 1: The 'Harder subset' plots (right two) have a y-axis scale (0.0–10.0) that is an order of magnitude smaller than the 'Easier subset' plots (0–60), which may obscure performance differences or make the 'Harder' results appear deceptively flat without explicit visual emphasis.
- **[writing]** Figure 3: The legend in the 'Repository Snapshot' panel uses the term 'core' to label the green file icons, but the caption defines 'core' as a 'span' (line range). This creates ambiguity between file-level and line-level granularity.
- **[writing]** Figure 3: The 'Explorer Output P' table uses the term 'core hit' in the 'GT label' column, but the legend in the 'Repository Snapshot' panel only defines 'core' and 'target', lacking a definition for 'hit'.
- **[fatal]** Figure 6: The caption is explicitly '(no caption)', yet the figure displays specific data (language distribution of 848 instances) that is identical to Figure 4. This renders the figure uninterpretable in isolation and suggests a duplicate file or missing caption error.
- **[science]** Figure 6: The chart displays a total of 848 instances, but the sum of the individual language counts shown (547+84+51+31+30+28+27+22+21+7) equals 848. However, the visual representation of the smallest slices (C++, C, Ruby) is extremely thin and crowded, making it difficult to distinguish them without the leader lines, which is a common issue with donut charts for many categories.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The manuscript relies heavily on specialized terminology that assumes a high degree of familiarity with the current landscape of LLM-based software engineering agents. While the target audience is likely researchers in this specific sub-field, several terms are used without definition, potentially excluding readers from adjacent fields (e.g., traditional software engineering or general NLP).

First, the term “**agentic**” is used repeatedly as both an adjective and a noun (e.g., “agentic explorers,” “agentic localization,” “agentic behavior”) without a definition. It is unclear if this refers to any system with an LLM, systems with tool-use capabilities, or a specific architectural

pattern. A brief clarification or replacement with “agent-based” or “autonomous” would improve clarity.

Second, **“trajectory-grounded”** and **“trajectory-grounded supervision”** are central to the paper’s methodology but are not explicitly defined in plain language. The text assumes the reader understands that this means deriving labels from the sequence of actions taken by a successful agent. A phrase like “ground truth derived from the step-by-step actions of successful agents” would be more accessible.

Third, the phrase **“downstream repair behavior”** appears frequently. While “downstream” is common in ML pipelines, “repair behavior” is slightly jargon-heavy. Replacing this with “subsequent code-fixing performance” or “ability to fix the bug” would be more direct. Similarly, **“restricted-context environment”** is used to describe the validation protocol but lacks a brief explanatory clause for what “restricted” entails (i.e., hiding parts of the codebase).

Finally, the term **“operating point”** in Section 5.2 (“occupy nearly the same operating point”) is technical jargon. Replacing this with “perform similarly” or “achieve comparable results” would make the finding clearer to a broader audience.

These changes are minor but necessary to ensure the paper’s contributions are accessible to the wider software engineering community, not just those deeply embedded in the current “agent” literature.

Action items.

- **[writing]** Define ‘agentic explorers’ and ‘agentic localization’ upon first use. The term ‘agentic’ is used as a noun/adjective without definition, assuming reader familiarity with specific agent architectures.
- **[writing]** Define ‘trajectory-grounded’ and ‘trajectory-grounded supervision’ at first mention. While context implies it relates to agent execution logs, the specific mechanism of deriving ground truth from trajectories needs a plain-language gloss.
- **[writing]** Replace ‘downstream repair behavior’ with ‘subsequent code-fixing performance’ or similar. ‘Downstream’ is standard in ML but can be opaque to general software engineering readers; ‘repair behavior’ is slightly jargon-heavy compared to ‘fixing bugs’.
- **[writing]** Define ‘restricted-context environment’ or ‘restricted-context validation’ clearly. The term implies a specific experimental setup but lacks a brief explanatory clause for non-specialists.
- **[writing]** Clarify ‘operating point’ in Section 5.2. The phrase ‘occupy nearly the same operating point’ is technical jargon from systems/ML that should be replaced with ‘perform similarly’ or ‘achieve comparable results’ for broader accessibility.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

“The paper presents a coherent logical framework for isolating repository exploration from patch synthesis. The definition of the task (ranked line-level region selection) follows logically from the identified gap in existing holistic benchmarks. The derivation of ground truth from intersecting successful trajectories is a sound methodological choice for creating a proxy for “necessary context,” provided the selection bias is acknowledged.

However, there are logical gaps in the interpretation of the correlation results. The abstract states: “This high correlation is expected: exploration metrics and downstream repair behavior both measure the same underlying capability.” This is a non-sequitur. High correlation indicates a strong relationship, not identity of measurement. The downstream protocol is designed to *validate* that exploration quality predicts repair success; it does not prove they measure the *same* capability. Repair success also depends on reasoning and code generation, which are distinct from exploration. The argument should be refined to state that exploration is a *limiting factor* or *necessary condition* for repair, rather than measuring the same capability.

Additionally, the selection criterion for ground truth (requiring ≥ 2 successful trajectories) creates a logical boundary that the paper does not fully address. The benchmark effectively evaluates exploration on “consensus-solvable” problems. It logically cannot make claims about exploration on problems where agents fail or where successful trajectories are idiosyncratic. The conclusion that “current agents are strong at finding relevant files but remain recall-limited” is valid only within this filtered subset. The paper should explicitly state that the findings do not generalize to instances where no consensus ground truth exists.

Finally, the explanation for the performance dip at $\alpha=25$ in the degradation analysis (Section 5.4) relies on a “memorization” hypothesis. While plausible, the paper presents this as a “plausible explanation” without ruling out alternative logical causes (e.g., noise introduction at low signal levels). The causal claim linking the dip to memorization is weakly supported by the data presented.”

Action items.

- **[science]** The abstract claims exploration and repair measure the ‘same underlying capability’ due to high correlation. This conflates correlation with identity. Repair also requires generation; the logic should state exploration is a necessary precursor, not the same capability.
- **[science]** Ground truth requires ≥ 2 successful trajectories, excluding unsolvable or idiosyncratic cases. This limits the benchmark’s logical scope to ‘consensus-solvable’ instances, contradicting claims of general repository exploration evaluation.
- **[writing]** The degradation analysis attributes a performance dip at $\alpha=25$ to ‘memorization’ without ruling out noise effects. This causal claim is speculative and unsupported by the restricted-context mechanism described.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper makes several claims that extend beyond the immediate evidence provided by the benchmark design and experimental results, specifically regarding the generalizability of the “agentic” advantage and the independence of the exploration metric from the repair outcome.

First, the abstract and introduction assert that “agentic explorers form a clear tier above classical retrieval.” While Table 2 confirms that agentic agents (e.g., AweAgent, LocAgent) significantly outperform lexical baselines (BM25, TF-IDF), the paper groups “classical retrieval” broadly. However, the dense retriever “Potion” (a non-agentic method) performs almost indistinguishably from the Random baseline (Rec₁ 0.025 vs 0.004). By grouping Potion with the successful agentic tier or failing to explicitly distinguish the failure of *dense* retrieval from *lexical* retrieval, the paper overstates the necessity of “agentic” behavior. The data supports that *interactive* or *reasoning-based* agents are superior to *lexical* search, but not necessarily that all non-agentic methods (like dense retrieval) are ineffective. This distinction is crucial for the field’s direction.

Second, the paper claims to “isolate repository exploration” from patch generation. However, the ground truth is constructed entirely from trajectories of agents that *successfully solved* the issue. This introduces a fundamental selection bias: the benchmark only measures exploration for problems that are solvable by current strong models. It cannot evaluate exploration for unsolvable issues or issues where the agent fails due to poor exploration but the code is actually present. The authors acknowledge this in the “Selection Bias Limitation” in the abstract but then proceed to claim the benchmark evaluates “repository understanding” and “bug diagnosis” in a general sense. The scope of these claims should be restricted to “exploration for successful repair” to avoid overclaiming the benchmark’s ability to diagnose general agent capabilities on the full spectrum of software issues.

Finally, the interpretation of the high correlation ($r=0.950$) between exploration metrics and downstream repair rates (Table 3) is slightly overreached. The authors state, “This high correlation is expected: exploration metrics and downstream repair behavior both measure the same underlying capability.” This phrasing suggests a tautological relationship or a universal truth. However, the downstream validation uses a *specific* fixed agent (Mini-SWE-Agent) and a *specific* context budget. The high correlation likely reflects the sensitivity of *that specific agent* to the provided context, rather than a universal property of exploration metrics. The paper should clarify that the metrics are predictive of repair *for the specific downstream protocol used*, rather than implying they measure a universal “repair capability” independent of the downstream agent’s architecture.

Action items.

- **[science]** The claim that ‘agentic explorers form a clear tier above classical retrieval’ overgeneralizes. Table 2 shows dense retriever ‘Potion’ fails like random, while only lexical baselines fail. The paper must clarify that the advantage is specific to agentic/interactive methods over lexical search, not all non-agentic methods.
- **[science]** The paper claims to ‘isolate’ exploration, yet ground truth is derived only from

successful repair trajectories. This biases the benchmark toward solvable issues. The authors admit this in the abstract but then claim to measure general ‘repository understanding.’ Claims should be narrowed to ‘exploration for successful repair’ to avoid overreach.

- **[writing]** The statement that high correlation ($r=0.950$) is ‘expected’ because metrics measure the ‘same underlying capability’ is an over-interpretation. The correlation is specific to the fixed downstream agent (Mini-SWE-Agent) and budget. The paper should clarify that metrics predict repair only within this specific protocol, not universally.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

The manuscript addresses safety and ethics primarily through a brief “Ethics Statement” and a single sentence in the Appendix regarding “Dual-Use Considerations.” While the authors correctly identify that the data comes from public repositories and is released under a permissive license, the current treatment of safety risks is insufficient for a benchmark explicitly designed to evaluate automated code analysis and exploration capabilities.

Dual-Use Risks: The benchmark isolates the “exploration” phase of software engineering, which is a critical precursor to both bug fixing and vulnerability discovery. As noted in the Appendix, there are dual-use implications. However, the current text (“Recommended access controls and monitoring”) is too vague. The review requires a more robust discussion on how this benchmark could be misused to automate the discovery of zero-day vulnerabilities or to generate targeted supply chain attacks. The authors should elaborate on the specific safeguards they recommend for users of the benchmark and whether they intend to implement any access controls on the dataset itself (e.g., requiring registration or usage agreements) to prevent malicious actors from using it to train vulnerability scanners.

Data Privacy and Sanitization: The ground truth is derived from agent trajectories that read code files. While the source repositories are public, the trajectories themselves might have inadvertently exposed sensitive information (e.g., API keys, internal IP addresses, or PII) if the agents were not perfectly constrained or if the repositories contained such data. The “Ethics Statement” mentions data was “sanitized,” but the methodology (Section 3.3 and Appendix B) lacks detail on *how* this sanitization was performed. Did the authors manually audit the trajectories? Did they use automated tools to scrub secrets? Without this detail, there is a risk that the released benchmark could inadvertently leak sensitive information that was present in the original agent logs.

Selection Bias and Societal Impact: The paper acknowledges a “Selection Bias Limitation” where only instances with at least two successful trajectories are included. From an ethics perspective, this creates a dataset that only represents “solvable” problems. If future agents are trained or evaluated primarily on this benchmark, they may become optimized for finding easy-to-

locate bugs while failing to address more complex, critical, or obscure vulnerabilities. The authors should briefly discuss this potential societal impact in the ethics statement, acknowledging that the benchmark’s design might skew the development of coding agents toward less critical issues.

The paper is generally sound in its scientific approach, but the safety and ethics sections require expansion to meet the standards expected for a benchmark with significant dual-use potential.

Action items.

- **[writing]** The ‘Dual-Use Considerations’ section (Appendix) is currently a single sentence recommending access controls. Given the benchmark’s focus on automated code analysis and vulnerability discovery, expand this to explicitly detail the specific risks (e.g., automated vulnerability scanning, supply chain attack generation) and the concrete mitigation strategies implemented (e.g., data sanitization protocols, release restrictions).
- **[writing]** The ground truth derivation relies on trajectories from proprietary models (GPT-5.4, Gemini-3-Pro, etc.). While the data is public, the methodology section should explicitly state whether any PII, secrets, or sensitive configuration data were present in the original trajectories and describe the specific sanitization process used to remove them before benchmark release.
- **[writing]** The selection bias limitation (requiring two successful trajectories) is acknowledged, but the ethics statement should briefly address whether this exclusion of unsolvable/complex issues might inadvertently bias the development of agents toward only ‘easy’ vulnerabilities, potentially leaving critical, hard-to-find bugs unaddressed by future tools trained on this data.

Scientific Evidence

Weights the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The paper presents a novel benchmark for repository exploration, but the scientific evidence supporting the generalizability of its findings requires clarification in three areas.

First, the ground-truth annotation methodology (Section 3.3) introduces a significant selection bias. By filtering for instances with at least two successful agent trajectories, the benchmark inherently excludes “unsolvable” or extremely difficult issues. Consequently, the reported metrics (e.g., line-level recall) reflect the exploration capabilities required only for problems that are already within the reach of current SOTA agents. The paper must explicitly address how this limitation affects the validity of claims regarding the “exploration bottleneck” for the broader set of software engineering tasks, particularly those that remain unsolved.

Second, the statistical robustness of the downstream validation (Section 5.2) relies on a subset of n=150 instances to compute correlations between exploration metrics and repair rates. While

the authors report 95% confidence intervals, the manuscript does not sufficiently justify the representativeness of this subset relative to the full 848-instance dataset. If the subset was selected based on difficulty or language distribution, the high correlations (e.g., $r=0.950$ for Context Efficiency) might not generalize to the full benchmark. A brief analysis of the subset’s distribution compared to the full set is necessary to rule out sampling bias.

Third, the controlled context degradation analysis (Section 5.4) concludes that missing context is the dominant failure mode based on a sharp performance jump between 50% and 75% of core evidence. While the trend is clear, the paper should provide statistical evidence (e.g., p-values for the pairwise comparisons mentioned in the abstract) confirming that this threshold effect is consistent across different patcher models and not an artifact of the specific GPT-5.4-mini configuration used. Without this, the claim that “modern patchers can tolerate extra irrelevant code” remains a strong observation rather than a statistically verified conclusion.

Action items.

- **[science]** The correlation analysis in Table 3 (Section 5.2) relies on a subset of $n=150$ instances. While CIs are now reported, the paper must explicitly justify why this subset is representative of the full 848-instance benchmark to avoid selection bias in the correlation claims.
- **[science]** The ground-truth construction requires at least two successful agent trajectories (Section 3.3). This creates a ‘survivorship bias’ where the benchmark only evaluates exploration on problems solvable by current SOTA agents. The authors must discuss how this limits the benchmark’s utility for evaluating explorers on harder, currently unsolvable issues.
- **[science]** In the degradation analysis (Section 5.4, Fig 3), the claim that ‘missing context is the dominant failure mode’ relies on a threshold effect observed in the data. The authors should clarify if the statistical significance of the jump between $\alpha=50$ and $\alpha=75$ holds across all tested patchers, or if it is specific to the GPT-5.4-mini/stronger pair.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The statistical analysis in SWE-Explore is generally sound, particularly in the use of trajectory-grounded ground truth and the downstream validation protocol. However, several critical statistical details are missing or insufficiently reported, preventing a full assessment of the robustness of the claims.

First, regarding the correlation analysis in Section 5.2 and Table 3: The authors report a very high Pearson correlation ($r = 0.950$) between Context Efficiency and downstream resolve rate based on a subset of $n = 150$ instances. While the magnitude is impressive, the absence of 95% confidence intervals (CIs) for both Pearson’s r and Spearman’s ρ is a significant omission. Given

the sample size, the precision of these estimates should be quantified. The abstract mentions that CIs are now reported, but they are not visible in the provided text or tables. Please ensure the final manuscript includes these intervals (e.g., via bootstrapping or Fisher’s z-transformation) to demonstrate that the correlation is not an artifact of the specific subset chosen.

Second, the degradation analysis in Section 5.4 and Figure 3 involves multiple hypothesis tests (pairwise comparisons across different α levels). The authors state that Holm-Bonferroni correction was applied, which is appropriate for controlling the family-wise error rate. However, the specific p-values and the underlying test statistics (e.g., t-statistics for the two-sample t-tests) are not reported in the text or the figure captions. Without these values, it is impossible to verify the significance of the reported “threshold-like pattern” or the claim that the dip at $\alpha = 25$ is a “caveat rather than the main effect.” Please add a supplementary table or expand the text to include the exact p-values and test statistics for the key comparisons (e.g., $\alpha = 50$ vs $\alpha = 75$).

Third, the ground-truth annotation methodology (Section 3.3) introduces a potential selection bias. The benchmark only includes instances where at least two successful agent trajectories were observed. This inherently filters out difficult or unsolvable issues. While the authors acknowledge this as a “Selection Bias Limitation” in the abstract, they do not quantify the extent of this filtering. How many instances were excluded from the original source datasets (SWE-bench Verified, Pro, Multilingual) due to this criterion? A quantitative analysis of the excluded set (e.g., comparing the difficulty or characteristics of excluded vs. included instances) is necessary to properly contextualize the claim that “missing context is the dominant failure mode.” If the excluded instances are those where context is irrelevant (because the issue is unsolvable regardless of context), the conclusion may be biased.

Finally, the use of $K = 5$ regions is justified by the average number of core regions being 4.7. However, the distribution of core region counts is not provided. If the distribution is highly skewed, a fixed $K = 5$ might be suboptimal for a significant portion of instances. A brief discussion or histogram of the core region count distribution would strengthen the justification for this hyperparameter choice.

In summary, while the experimental design is innovative, the statistical reporting requires enhancement to meet the rigor expected for a benchmark paper. Specifically, the addition of confidence intervals, full statistical test results, and a quantitative analysis of the selection bias is required.

Action items.

- **[science]** The abstract and Section 5.2 report a Pearson correlation of $r=0.950$ on $n=150$ instances but omit 95% confidence intervals. Please report the CI for both Pearson’s r and Spearman’s ρ in Table 3 to quantify the precision of these estimates.
- **[science]** In Section 5.4 (Degradation Analysis), pairwise comparisons (e.g., $\alpha=50$ vs $\alpha=75$) are mentioned with Holm-Bonferroni correction, but the specific p-values and the exact test statistic (t-statistic) are not reported in the text or tables. Please add these values to ensure reproducibility.
- **[science]** The ground-truth construction relies on the intersection of successful trajectories (Section 3.3). This introduces a selection bias where only solvable instances are included.

Please explicitly quantify the size of the excluded set (instances with <2 successful trajectories) and discuss how this limits the generalizability of the ‘missing context is dominant’ conclusion to unsolvable or hard-to-solve issues.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The manuscript presents a clear and well-structured argument for the SWE-Explore benchmark. The writing is generally professional, with a logical flow from the motivation in the Introduction to the detailed methodology and results. The separation of concerns between the benchmark definition and the downstream validation protocol is articulated effectively.

However, there are a few areas where the prose could be tightened to improve precision and tone. The Abstract includes a sentence (“This high correlation is expected...”) that sounds defensive and interrupts the objective reporting of results; this should be rephrased to maintain a neutral, analytical voice. In Section 3.2, a subject-verb agreement error (“Table 1 and Figure 1 summarizes”) disrupts the reading flow. Additionally, some interpretive statements in the Results section (e.g., Section 5.2) use vague phrasing like “larger pattern” which could be replaced with more specific descriptors to ensure the reader clearly understands the observed trends.

Overall, the paper is highly readable, but these minor stylistic and grammatical adjustments would elevate the quality of the writing to match the rigor of the research.

Action items.

- **[writing]** In the Abstract, the sentence ‘This high correlation is expected: exploration metrics and downstream repair behavior both measure the same underlying capability’ reads as a defensive justification rather than a neutral scientific observation. Rephrase to state the finding objectively (e.g., ‘This strong correlation suggests that...’) to maintain the paper’s analytical tone.
- **[writing]** Section 3.2 (Data Sources) contains a grammatical agreement error: ‘As Table 1 and Figure 1 summarizes...’ should be ‘summarize’ to agree with the plural subject. Additionally, the phrase ‘embedded in repositories that average 759 files’ is slightly clunky; consider ‘with repositories averaging 759 files’ for better flow.
- **[writing]** In Section 5.2, the phrase ‘The larger pattern is more important than the exact ordering’ is vague. Clarify what ‘larger pattern’ refers to (e.g., ‘The consistent gap between file-level and line-level performance’) to ensure the reader immediately grasps the intended takeaway without ambiguity.